CMPUT 461/561 Fall 2026

Lec 00 -- Overview and course policies

Overview of course content

- What is NLP about?
- Assistants
- pre-LLM versus LLM-based NLP
- Admin

What is NLP about?

Natural Language Processing (NLP) is about enabling computers to *understand*, interpret, and generate human language in a way that is both meaningful and useful.

- Bridge the gap between **human communication** and **computer understanding**.

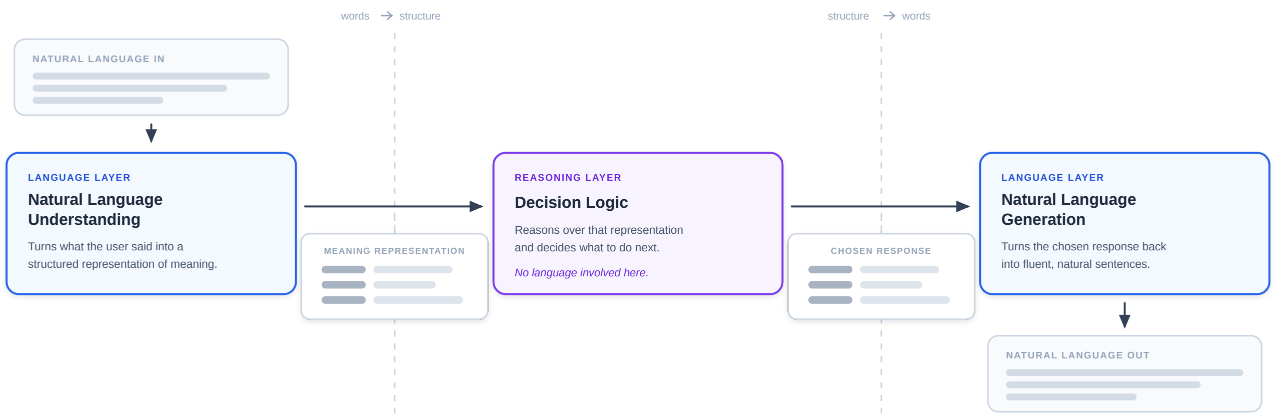
Human communication

- Ambiguous
- Contextual / implicit
- Emotional
- Continuously evolving

Computer understanding

- Literal / formal
- Explicit
- Rational
- Static

General AI assistant/chatbot architecture



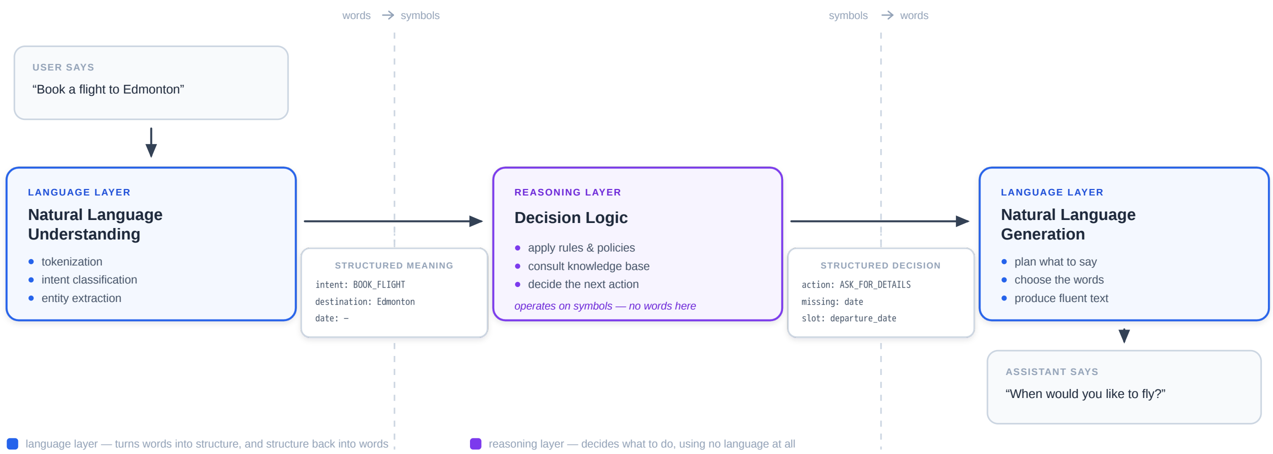
Understanding is hard

An AI agent dealing with a user request in natural language needs to solve many problems before taking any action.

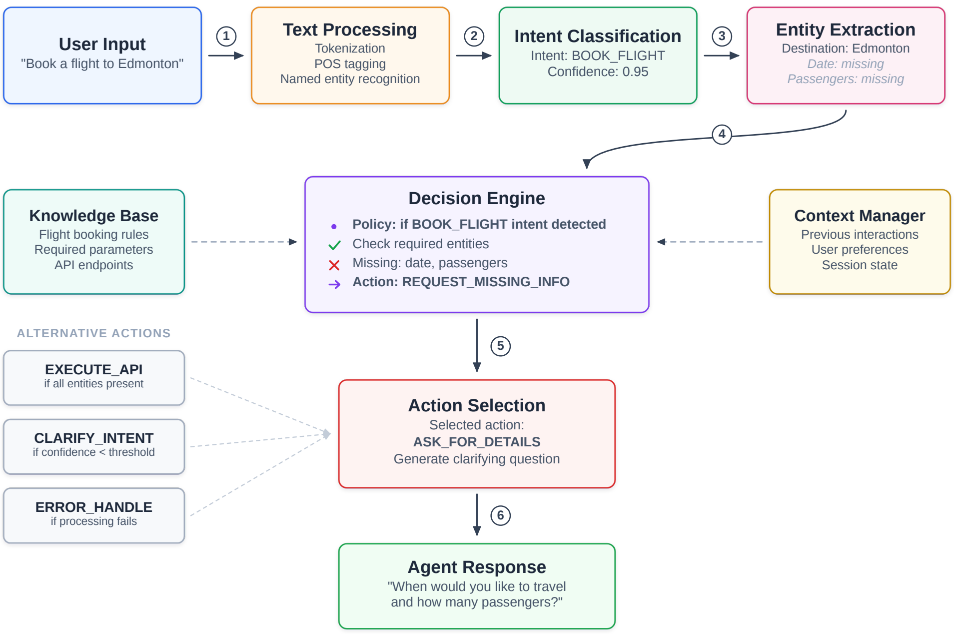
Example: What does "book" mean here?

This is an example of **lexical ambiguity** (when a word has multiple possible meanings).

AI travel agent booking a flight



A hypothetical pipeline



Typical NLP Tasks in Conversational Tools

Examples drawn from a travel-agent assistant.

Understanding the user

- **Language identification** — which language is the user speaking?
- **Spelling correction** — repair typos before matching ("Edmonten" → "Edmonton").
- **Intent classification** — what does the user want? `BOOK_FLIGHT` vs. `CANCEL_BOOKING`.
- **Named entity recognition (slot filling)** — pull out cities, dates, passenger counts.
- **Coreference resolution** — resolve "it", "that one", ... to an actual referent.
- **Semantic parsing** — map the whole utterance to a structured query or API call.
- **Sentiment / emotion detection** — spot frustration and escalate to a human.

Producing the reply

- **Natural language generation** — turn the chosen action into a sentence.
- **Summarization** — condense twelve flight options into a readable shortlist.
- **Machine translation** — reply in the user's language.

If the interface is voice

- **Speech recognition (ASR)** — audio to text on the way in.
- **Text-to-speech (TTS)** — text to audio on the way out.

Solutions to typical NLP tasks

Pre-LLM

Requires pre-processing (tokenization; tagging; entity recognition; etc.), often customized for the task.

Uses a custom algorithm or a learned model (requires specialized data).

Can be fine-tuned and deployed locally.

LLM-based

Zero shot: just ask an LLM to do it.

Prompt engineering: customize the LLM prompt for better performance.

Can fine-tune a model or build a tool (normally, a classifier) that uses an LLM (or some of its components).

Pre-LLM intent classification with rules

Pre-processing: tokenize the text; assign parts-of-speech to tokens.

Intent classification: determine what is it that the user wants.

Ambiguity: is `book` a verb (action) or a noun (object)?

If it is an action... is that an action the agent can perform?

And if so... is `flight` something the agent can apply that action to?

Ambiguity and understanding

Generally, we cannot disambiguate words individually.

Instead, the AI must "interpret" the entire sentence at once.

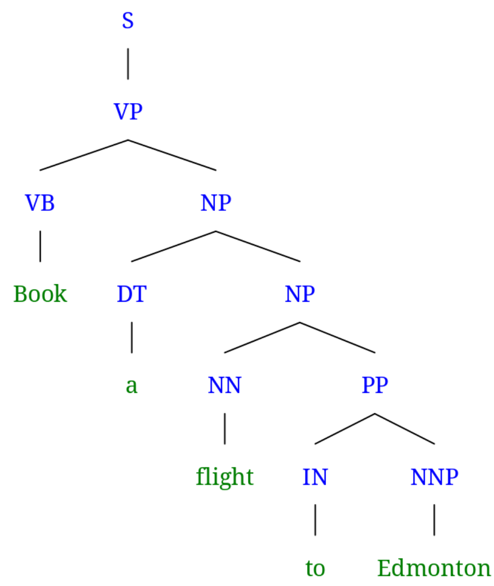
- Look for a combination of **parts of speech** that can be assigned to individual words
- Look for **grammar rules** that, together, can assign meaning to the whole sentence.

Dealing with ambiguity: parsing

"Classical" NLP methods were based on linguistic knowledge:

- The **imperative** is a grammatical mood that forms a command or request.
- **In English**, the imperative is formed using the infinitive form of the verb.

The token `book` corresponds to the infinitive form of a verb!

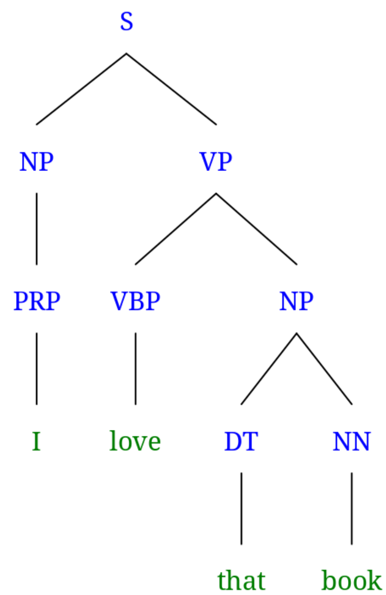


Dealing with ambiguity: parsing

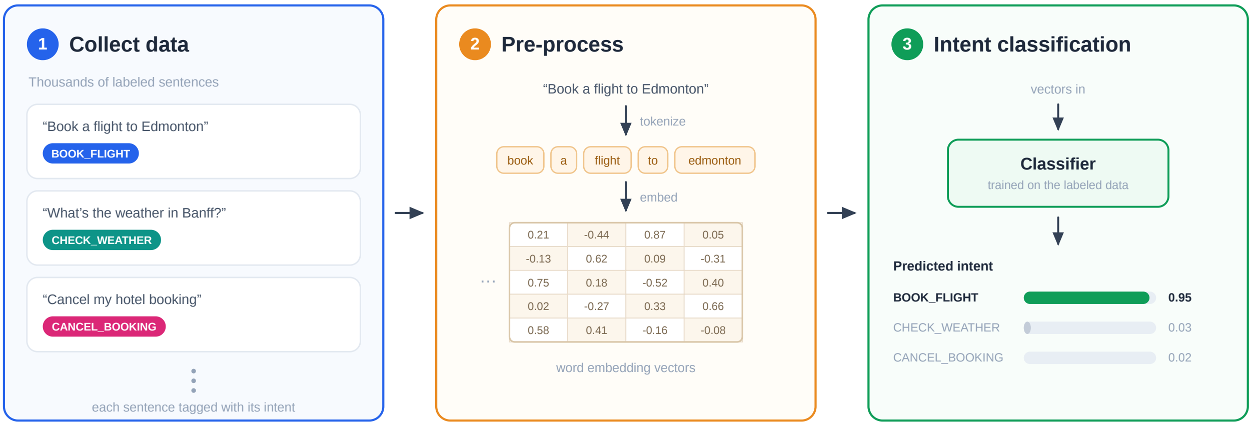
More linguistic knowledge:

- The **realis mood** (a.k.a. the *indicative*) is used to state facts.
- **In English**, the indicative can follow a subject-verb-object form.

subject: I **verb**: love **object**: "a" book



pre-LMM intent classification with machine learning



LLM-era intent classification

Zero-shot classification

D What is the intent of the request "Book a flight to Edmonton"?

The intent of the request "Book a flight to Edmonton" is to **make a travel reservation** - specifically to purchase or reserve airline tickets for travel to Edmonton (likely Edmonton, Alberta, Canada).

Just ask an LLM to detect intent in the user input!
No need for algorithms, models, training data, ...

From "Classical" to "modern" NLP

1960s-1990s: NLP methods relied strongly on Linguistics, mathematical formalisms and logical structures.

1990s-2010s: NLP methods relied increasingly on supervised machine learning.

2010s onwards: NLP now relies of models trained on massive datasets, out of reach of most except the wealthiest companies.

What was wrong with "classical" methods?

`TL;DR`: high complexity and cost; low performance

High expertise required

- Linguists and computer scientists earn high $$$.

Low scalability

- Systems require maintenance as language keeps evolving.

Underwhelming performance

- Natural language is very complex; systems rarely lived up to expectations.

Why are "modern" methods so popular?

`TL;DR`: easier to "program"; impressive results; hard to debunk

Machine learning is an easier "programming" paradigm, that requires no experts

- Collecting "good enough" training data is cheap and can be done by anyone.

Zero-shot "learning" seems like a free lunch, does it not?

Impressive performance on many tasks

- It is nearly impossible to debunk myths about superhuman performance.

Black box nature

- Many API flavours; easy to deploy in a plug-and-play fashion.

Has machine learning/zero-shot learning solved NLP?

Absolutely not!

NLP remains a challenging and rewarding field of active research and development.

New methods brought in **massive challenges** and opportunities

- Higher productivity (and mass unemployment?)
- Hallucination (it is inevitable!)
- How to detect a mistake? And what to do when that happens?

What will CMPUT461/561 cover?

① "Classical" Assistant and "modern" Conversational agent

② "Classical" and probabilistic methods for:

- Word recognition; part of speech tagging; parsing
  - Enough for you to understand how agents were built up to a few years ago

③ Probabilistic language models and intro to LLMs:

- Training and text generation
- Safety and ethical concerns

④ Contrasting "classical" and "modern"

CMPUT 461 Grading

Item	Weight	Date
test 1	25%	Oct 29
test 2	25%	Dec 3
final	45%	TBA
project	5%	see Canvas

Letter Grade Assignment

```
≥ 95% A+   ≥ 90% A    ≥ 85% A-
≥ 80% B+   ≥ 75% B    ≥ 70% B-
≥ 65% C+   ≥ 60% C    ≥ 55% C-
≥ 50% D+   < 50% F
```

CMPUT 461 Exams

	tests	final
Closed book?	✅	✅
Cheat sheets allowed?	❌	❌
Calculators allowed?	❌	❌
Any other aids allowed?	❌	❌
Smartphones and smartwatches allowed in the room?	❌	❌
Multiple-choice questions answered on scantron papers?	✅	✅
Questions about coding and numerical problems?	✅	✅
Open-ended questions or problems?	❌	✅

CMPUT 461 -- missed term work

Tests are eligible for **excused absences**:

- Must be requested following rules in the UfoA Calendar.
- Granted or denied based on rules in the Calendar
- If granted, student will write a **modified final exam** containing all questions in the final exam and the missed midterm exam(s).
  - No extra time.

Deferred final exams must be requested to the Faculty of Science

- Will be held on a date TBA, but not before the Winter break.

CMPUT 461 Project

Build a language model and a chatbot that can answer simple questions about a domain of your choice (ex: biology or geography).

Four parts:

1. Collect data from Wikipedia (1%)
2. Learn a model (1%)
3. Build a chatbot (1%)
4. Evaluate the system (2%)

Bonus marks (double) for each part, if TAs deem solution to be outstanding.

Rules:

1. Can use AI.
2. Cannot submit multiple "parts" in the same deadline.
3. Cannot skip any part (to submit part 3, must have done parts 1 and 2).
4. No extensions. No EAs.

CMPUT 561 Grading

Item	Weight	Date
test 1	20%	Oct 29
test 2	20%	Dec 3
essay	15%	Dec 8
assignments (5)	25%	see Canvas
reviews (5)	20%	see Canvas

Letter Grade Assignment

```
≥ 95% A+    ≥ 90% A    ≥ 85% A-
≥ 80% B+    ≥ 75% B    ≥ 70% B-
≥ 65% C+    ≥ 60% C    ≥ 55% C-
≥ 50% D+    < 50% F
```

CMPUT 561 -- missed term work

Tests and the essay are eligible for and **excused absence**:

- Must be requested following rules in the UfoA Calendar.
- Granted or denied based on rules in the Calendar
- If granted, student will write an exam at the same date/time/location and with the same duration of the **CMPUT461 final exam**.

No excused absences, substitutions or extensions for paper reviews and assignments.

CMPUT 561 -- the Reflection Essay

Step	Date	In person?
Write an essay (in English) by hand on a sheet of paper; take a picture of that paper, **hand it in**	December 8, 2025	✅
Type your essay up (from the picture)	Before Dec 15	❌
Complete the online assignment	Before Dec 15	❌

The essay must be based on course material.

The assignment asks you to obtain and compare a second essay, written by an LLM.

CMPUT 561 -- assignments

Assignment	Topic	code?	writing?
asn1	Regular expressions, tokenization, word recognition	✅	✅
asn2	Grammars	✅	✅
asn3	Probabilistic language modeling	✅	✅
asn4	Natural language generation	✅	✅
asn5	Part-of-speech tagging	✅	✅

Using AI is strongly advised.

Goal: understand how LLMs can help with coding tasks. Reflect on that.

CMPUT 561 -- paper reviews

Groups of papers are already selected for you.

Targeted reviews, with specific questions you must answer.

Paper review methodology will be provided.

Must submit annotated paper (PDF or printout) with notes that match your review.

CANNOT USE AI for the reviews.

CANNOT USE AI for writing assistance.

Email

Subject line **must** contain student number or CCID.

Questions about the content or course policies must go to `cmput461@ualberta.ca`.

Grievances with a TA and EA requests must be sent to the instructor (see Canvas).

Office hours

See Canvas

Every case of misconduct will be reported

There has been a noticeable increase in abusive language from students towards instructors and staff at the University of Alberta in recent years.

Using **abusive language** in emails, forum posts, or in person is a form of **harassment**.

"Harassment is a **single or repeated** incident of objectionable, unwelcome or adverse conduct, ..." (University of Alberta's Student Conduct Policy)

Every single case of harassment will be reported for investigation.